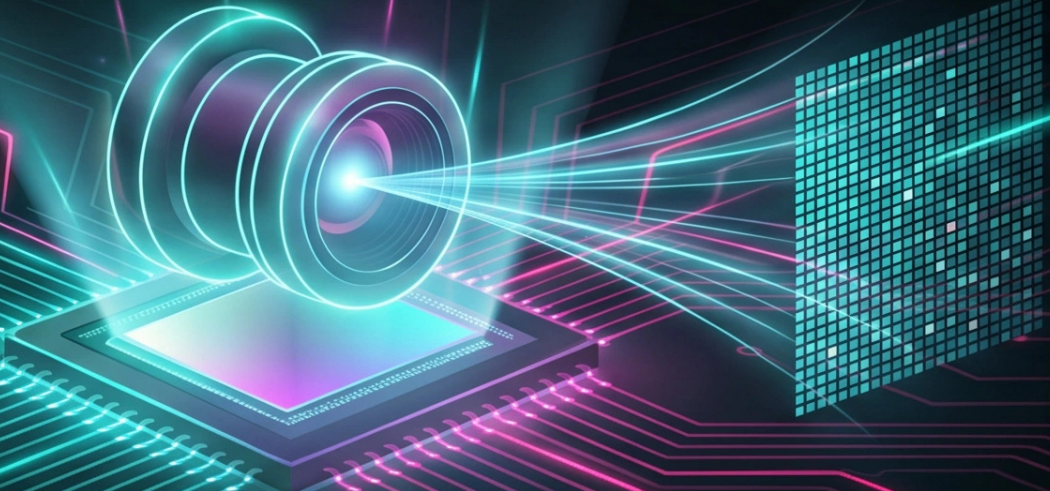


CAMERA SYSTEMS: FROM PHOTON TO PIXEL

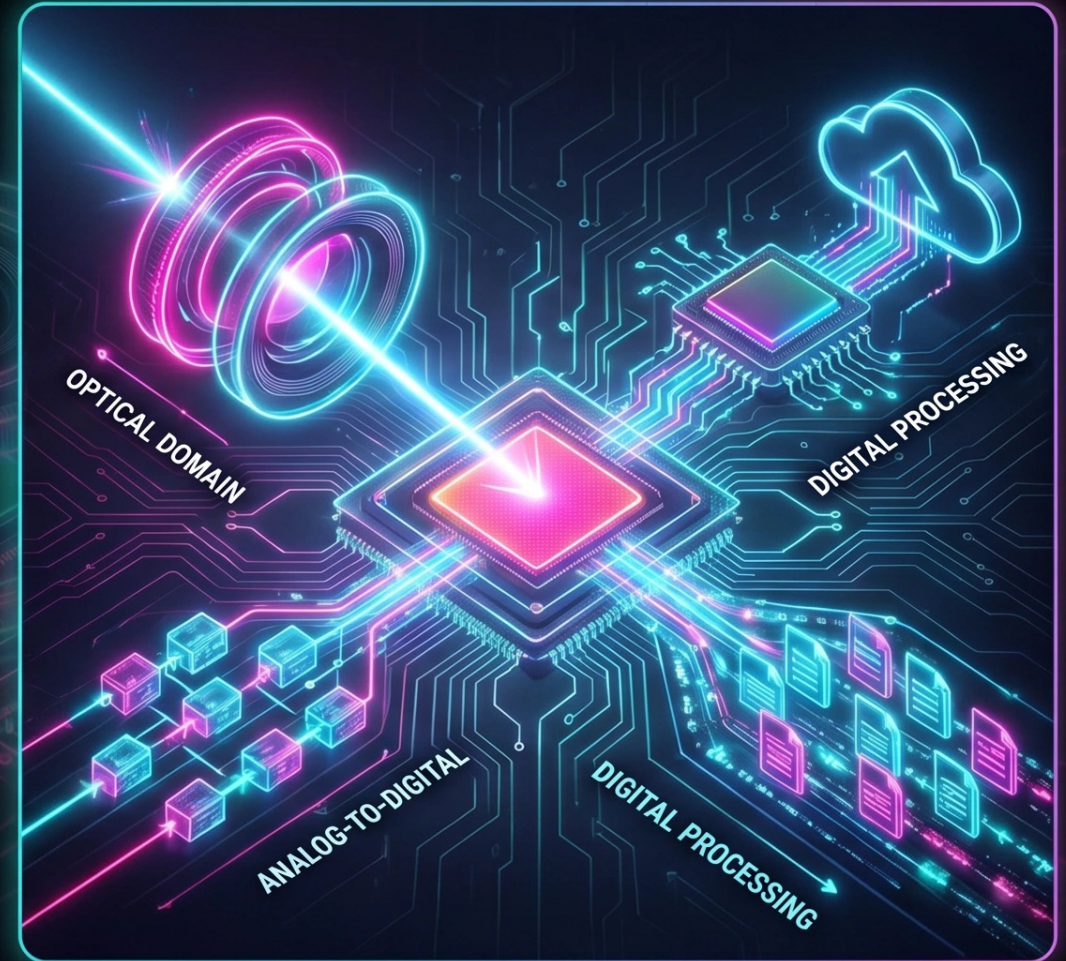
Understanding the Complete Camera Pipeline —
Sensor, ISP, SoC, and Software Stack



MURAT BALCI, PhD

THE JOURNEY OF LIGHT: CAMERA SYSTEM OVERVIEW

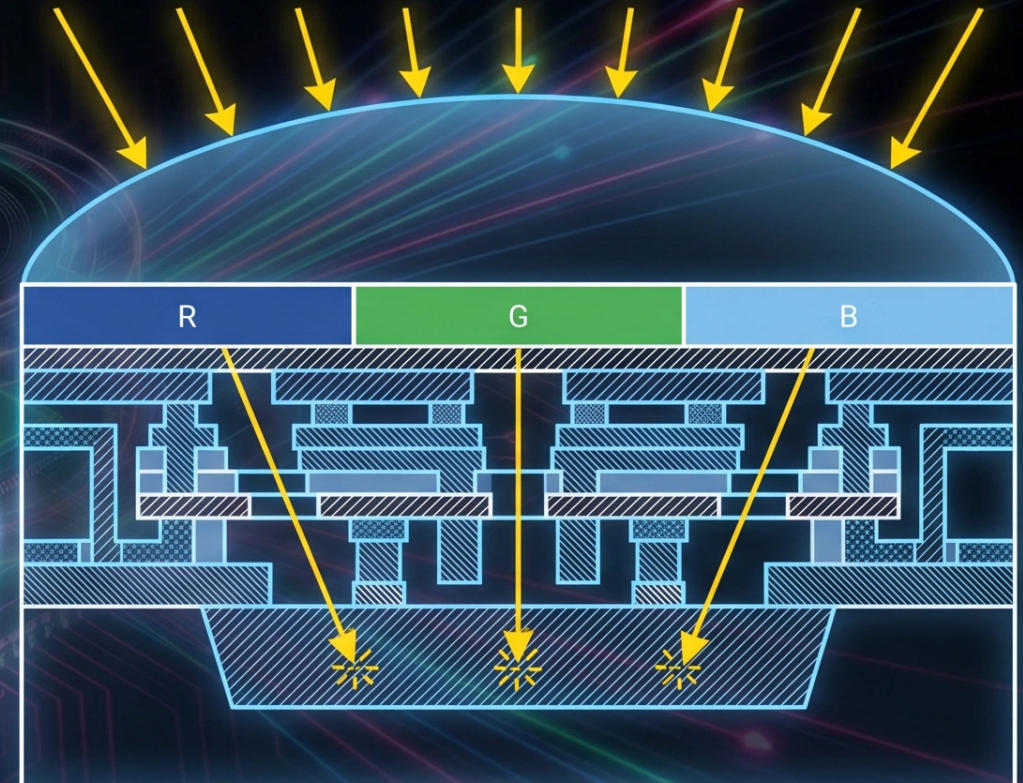
- 💡 • Light enters through the lens and strikes the image sensor, beginning a multi-stage transformation.
- 🔍 • The complete pipeline spans three domains: optical (lens and filters), analog-to-digital (sensor and ADC), and digital processing (ISP, SoC, software).
- 🖼️ • Every smartphone photo passes through 10+ processing stages in under 33 milliseconds.
- ⚙️ • Understanding this pipeline is essential for camera firmware, ISP tuning, and computational photography.



CMOS IMAGE SENSOR: ANATOMY OF A PIXEL

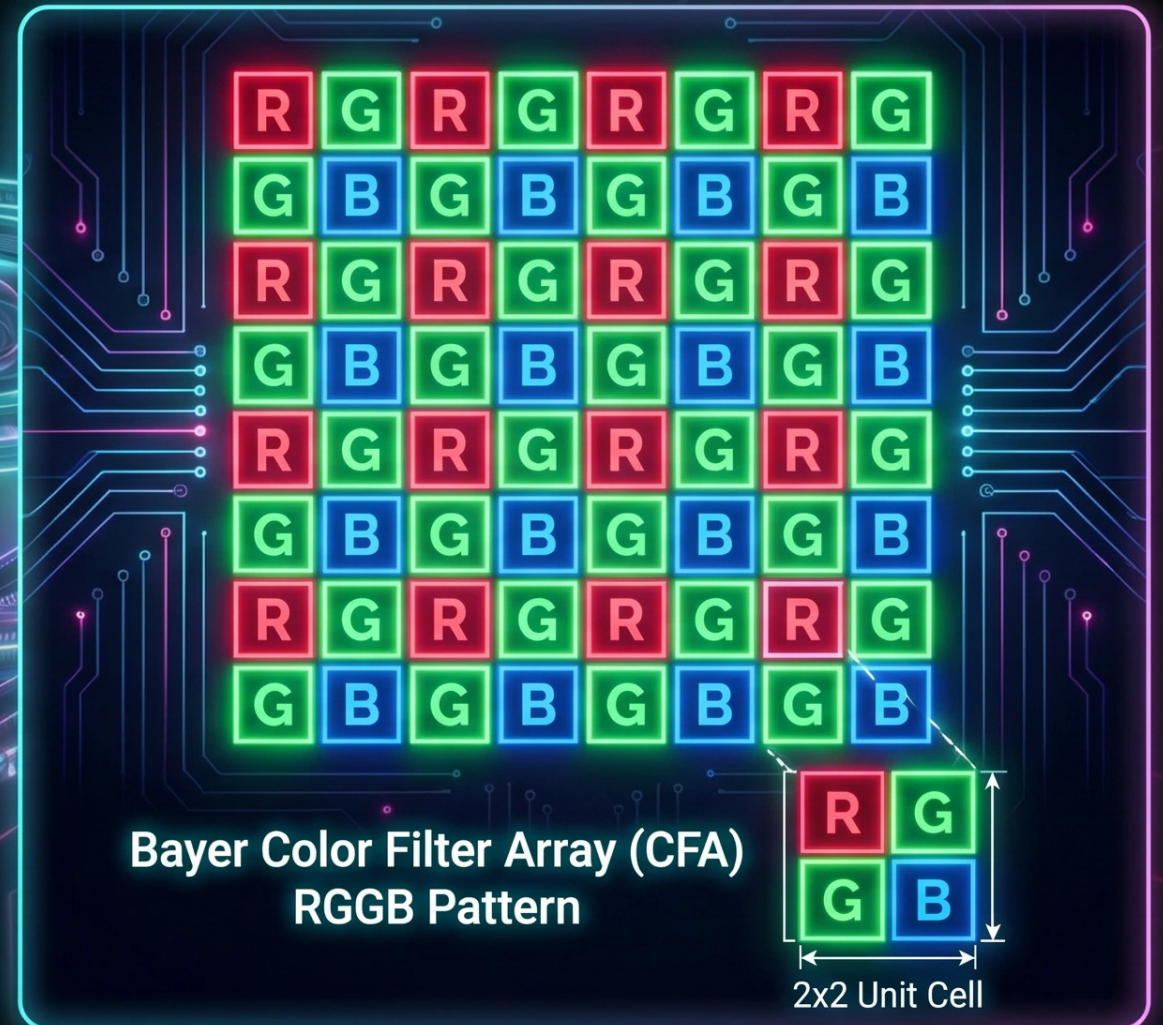
- KEY POINTS:

- 1. Photodiode converts photons to electrical charge.
- 2. Microlens focuses light, minimizing waste.
- 3. Bayer CFA overlays sensor (50% Green, 25% Red, 25% Blue).
- 4. ADC digitizes analog signal (10-14 bits/pixel).



BAYER FILTER MOSAIC: HOW COLOR IS CAPTURED

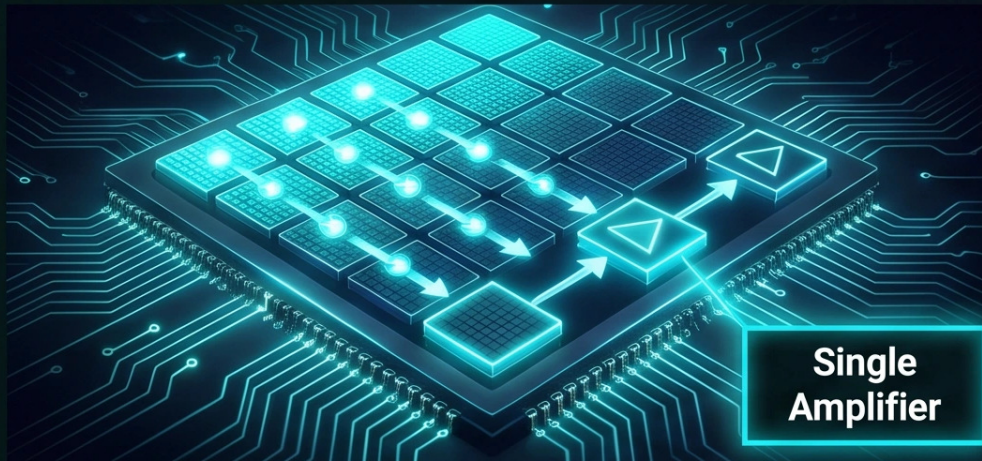
- Photodiodes are inherently color-blind (measure only grayscale intensity).
- Bayer CFA assigns one color per pixel (R, G, or B).
- RGGB 2x2 unit cell repeats across the sensor.
- Raw output is a single-channel mosaic, requiring demosaicing to reconstruct full RGB.



CCD vs. CMOS: Sensor Technology Comparison

Understanding the Fundamental Differences in Image Sensor Architecture

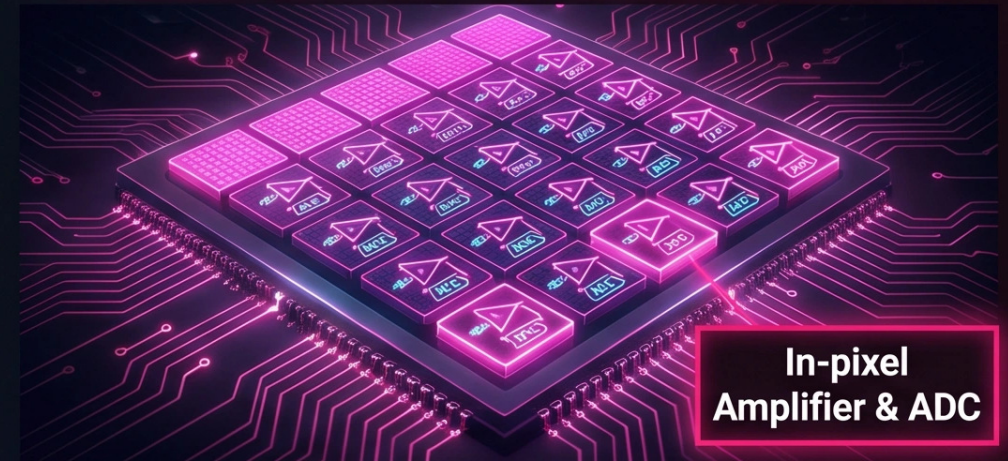
CCD: Charge-Coupled Device



Single Amplifier

- Shifts charge row-by-row to a single amplifier
- High image quality
- Higher power consumption
- Slower readout

CMOS: Complementary Metal-Oxide Semiconductor



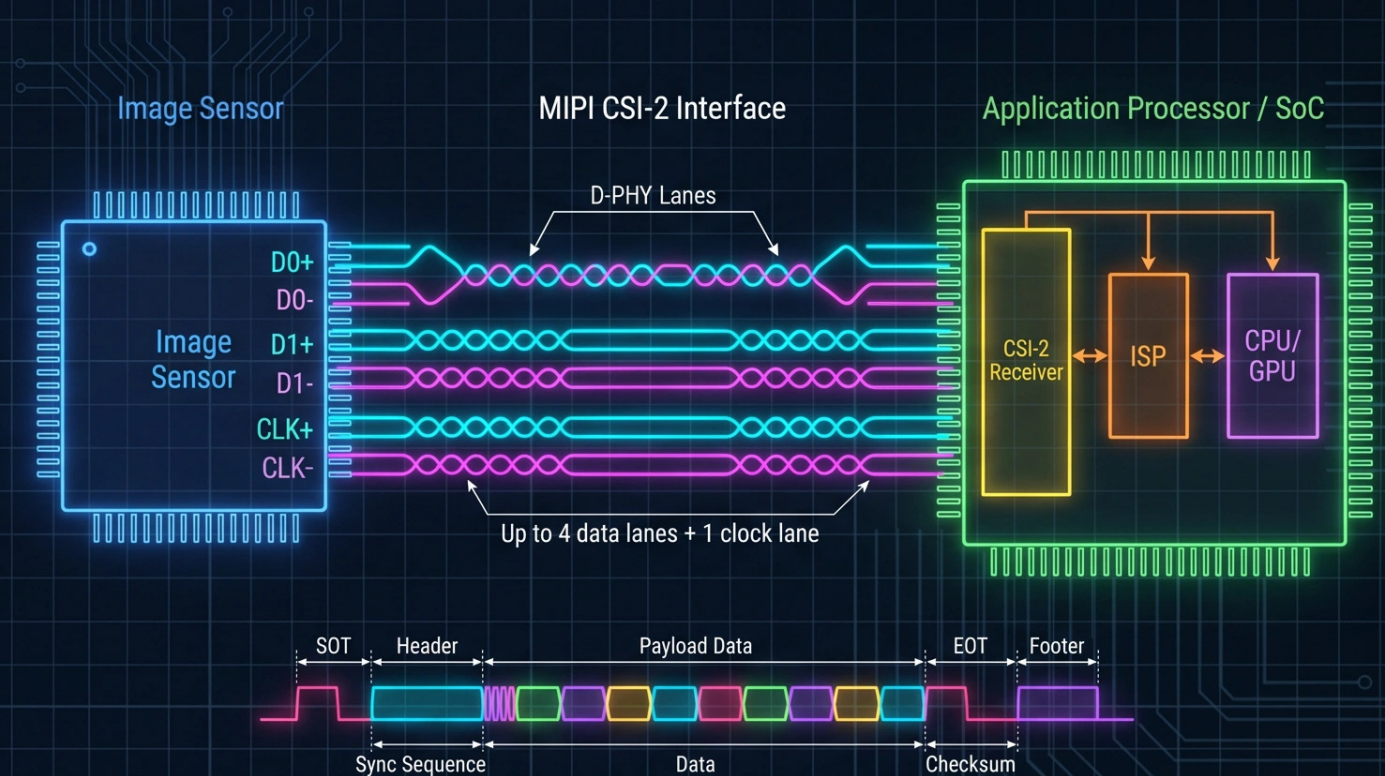
In-pixel Amplifier & ADC

- Each pixel has its own amplifier and ADC
- Faster readout, lower power
- Enables on-chip processing

Modern cameras overwhelmingly use CMOS sensors due to speed, integration, and cost advantages. CMOS enables features like rolling shutter, HDR pixel architectures, and on-sensor phase-detection AF.

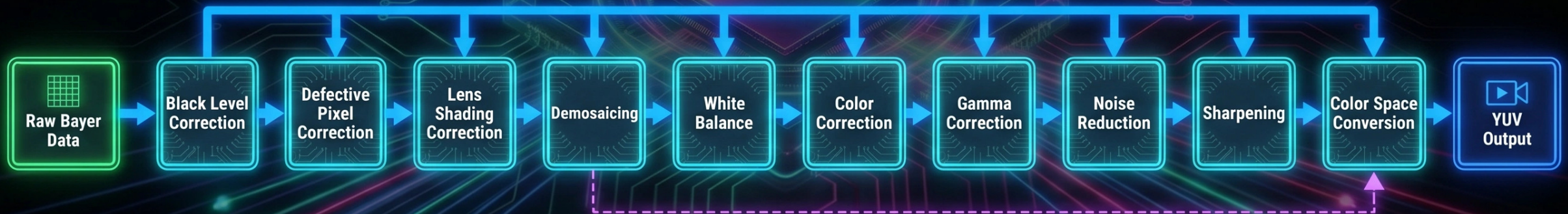
SENSOR TO ISP: THE MIPI CSI-2 INTERFACE

- Raw Bayer data transmitted via MIPI CSI-2 protocol.
- Uses D-PHY differential signaling (up to 4 data lanes + 1 clock lane).
- Data is packetized: SOT -> Header -> Pixel Payload -> Checksum -> EOT.
- SoC's CSI-2 Receiver deserializes data for the ISP.
- Sensor control uses a separate I2C/I3C bus.

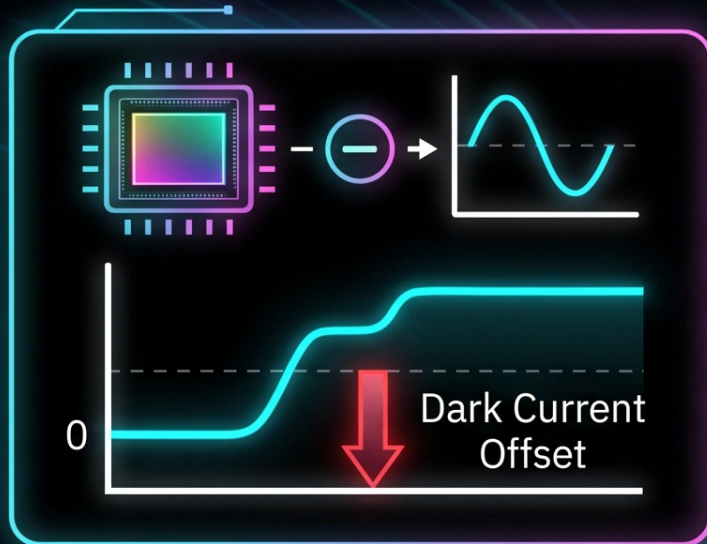


ISP Pipeline: The 10-Stage Image Factory

- Dedicated hardware block transforms raw Bayer data into a viewable image.
- Stages execute in fixed pipeline order in real time.
- Key stages: Black Level -> Demosaicing -> White Balance -> Color Correction -> Gamma -> Noise Reduction -> YUV Output.
- Outputs YUV data for encoding/display.

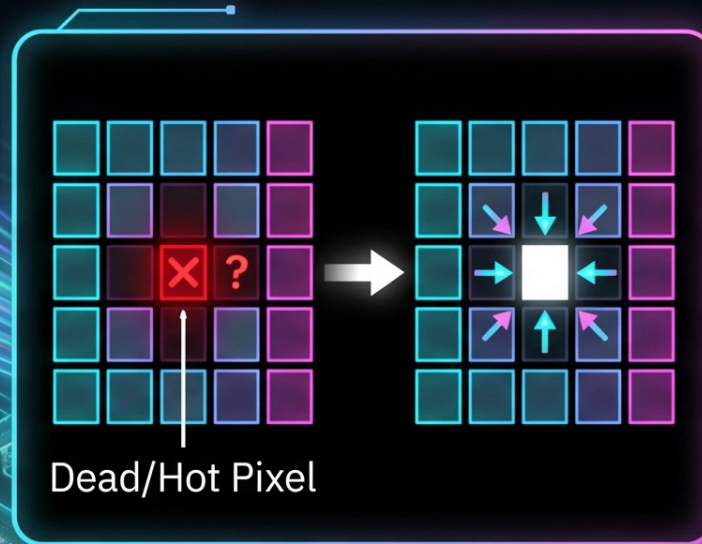


ISP STAGE 1-3: SENSOR CORRECTIONS



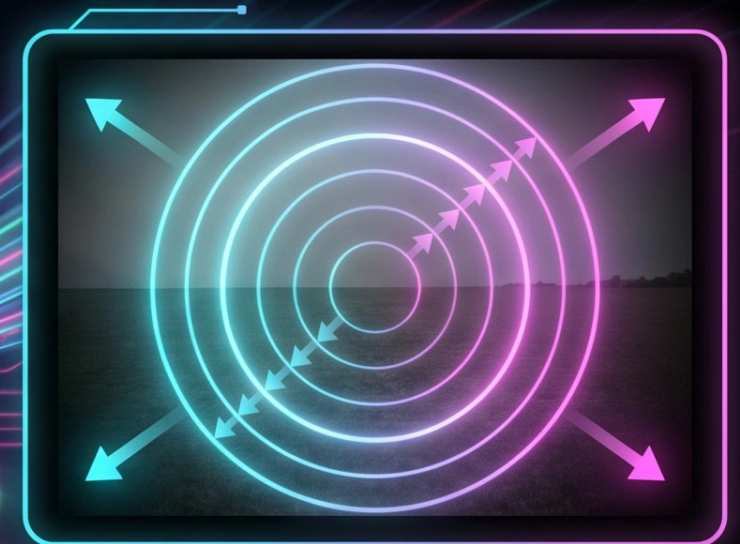
1. BLACK LEVEL CORRECTION

Subtracts the sensor's dark current offset (removes non-zero values when no light).



2. DEFECTIVE PIXEL CORRECTION

Identifies dead/hot pixels and replaces them with interpolated neighbor values.



3. LENS SHADING CORRECTION (VIGNETTING)

Compensates for light falloff at image edges by applying a radial gain map.

ISP Stage 4: Demosaicing – Reconstructing Full Color

- Key Points:
 - The most computationally intensive ISP stage: interpolates the two missing color channels for every pixel.
 - A green pixel knows its G value but must estimate R and B from surrounding pixels.
 - Algorithms range from simple bilinear interpolation to advanced adaptive edge-directed methods.
 - Poor demosaicing causes visible artifacts: color moiré, zipper artifacts, and false color at sharp edges.
 - Modern ISPs use hardware-accelerated adaptive algorithms that analyze local gradients.



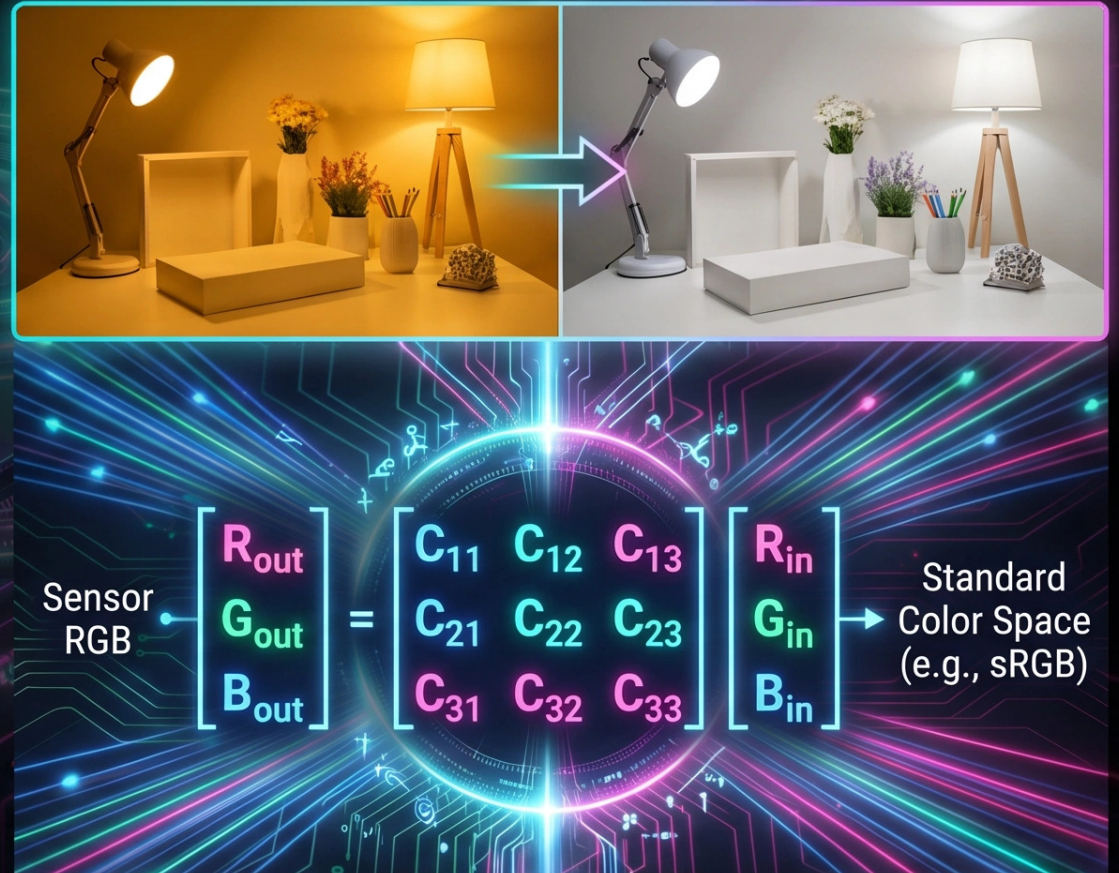
Interpolation Process: Central Green Pixel Receiving Red and Blue Data

ISP Stage 5-6: White Balance and Color Correction

Optimizing Image Color Accuracy & Fidelity

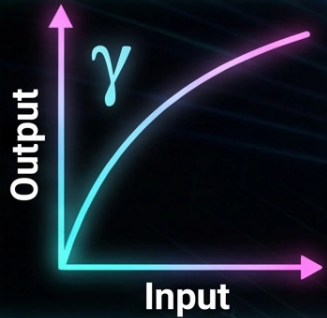
Key Points

- **1. Auto White Balance (AWB):** Removes color cast from scene illumination, making whites appear neutral regardless of light source.
- **2. AWB Algorithms:** Estimate the illuminant using methods like gray world assumption or white patch detection.
- **3. Color Correction Matrix (CCM):** A 3x3 matrix that maps sensor-specific RGB values to a standard color space (sRGB, DCI-P3).
- **4. CCM Compensation:** Compensates for differences between the sensor's spectral response and the human visual system.



ISP STAGE 7-10: TONE MAPPING, DENOISING, AND OUTPUT

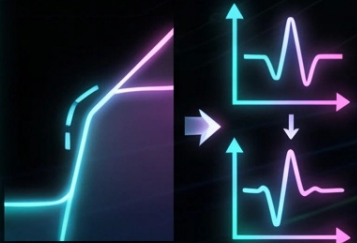
Finalizing Image Quality for Display and Efficient Storage



1. Gamma Correction: Applies non-linear transfer function (sRGB ~ 2.2) to match human perception.



2. Noise Reduction: Spatial and temporal filtering removes noise while preserving edges.



3. Sharpening: Enhances edge contrast to counteract softening.

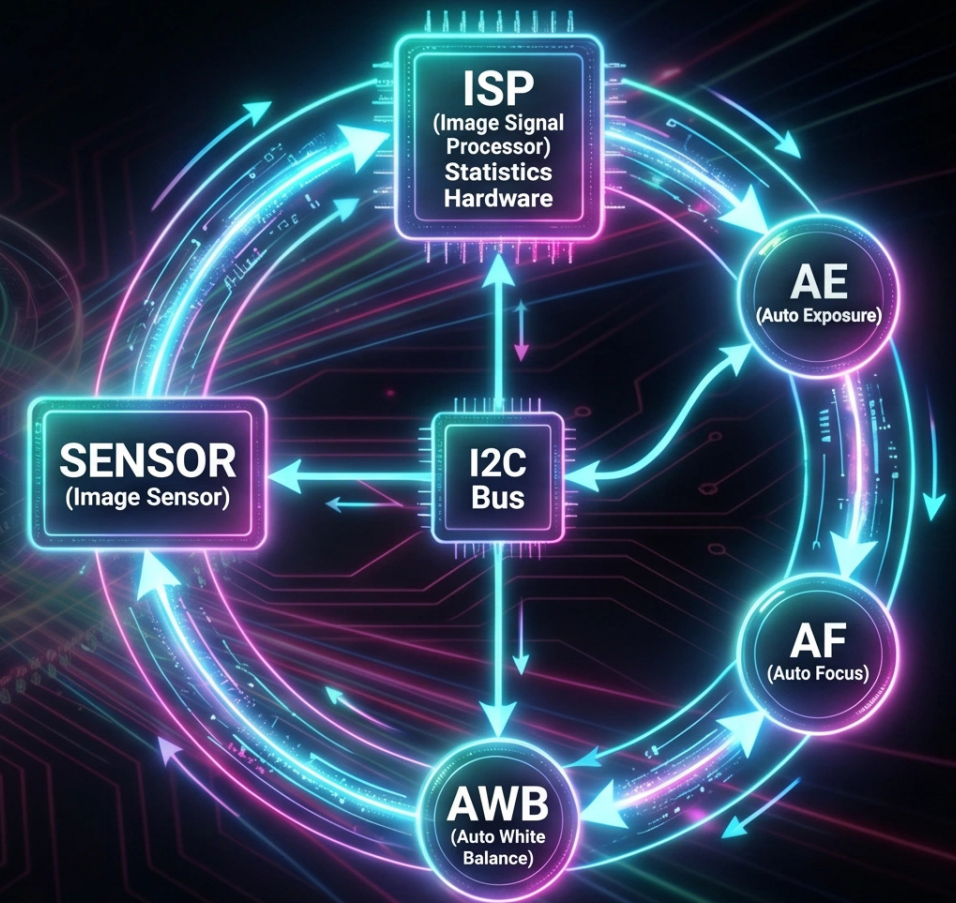


4. Color Space Conversion: Transforms RGB to YUV/YCbCr for efficient encoding.

Presenter: Engineering Lecture Series

The 3A Algorithms: Closed-Loop Sensor Control

- **1. Auto Exposure (AE):** Analyzes frame brightness statistics to adjust sensor exposure time and analog gain.
- **2. Auto Focus (AF):** Maximizes image sharpness by analyzing contrast or using phase-detection pixels.
- **3. Auto White Balance (AWB):** Continuously estimates scene illuminant and adjusts R/G/B channel gains.
- 4. 3A algorithms run on ISP statistics hardware, feeding control signals back to the sensor via I2C.
- 5. Convergence speed and stability directly impact user-perceived camera responsiveness.

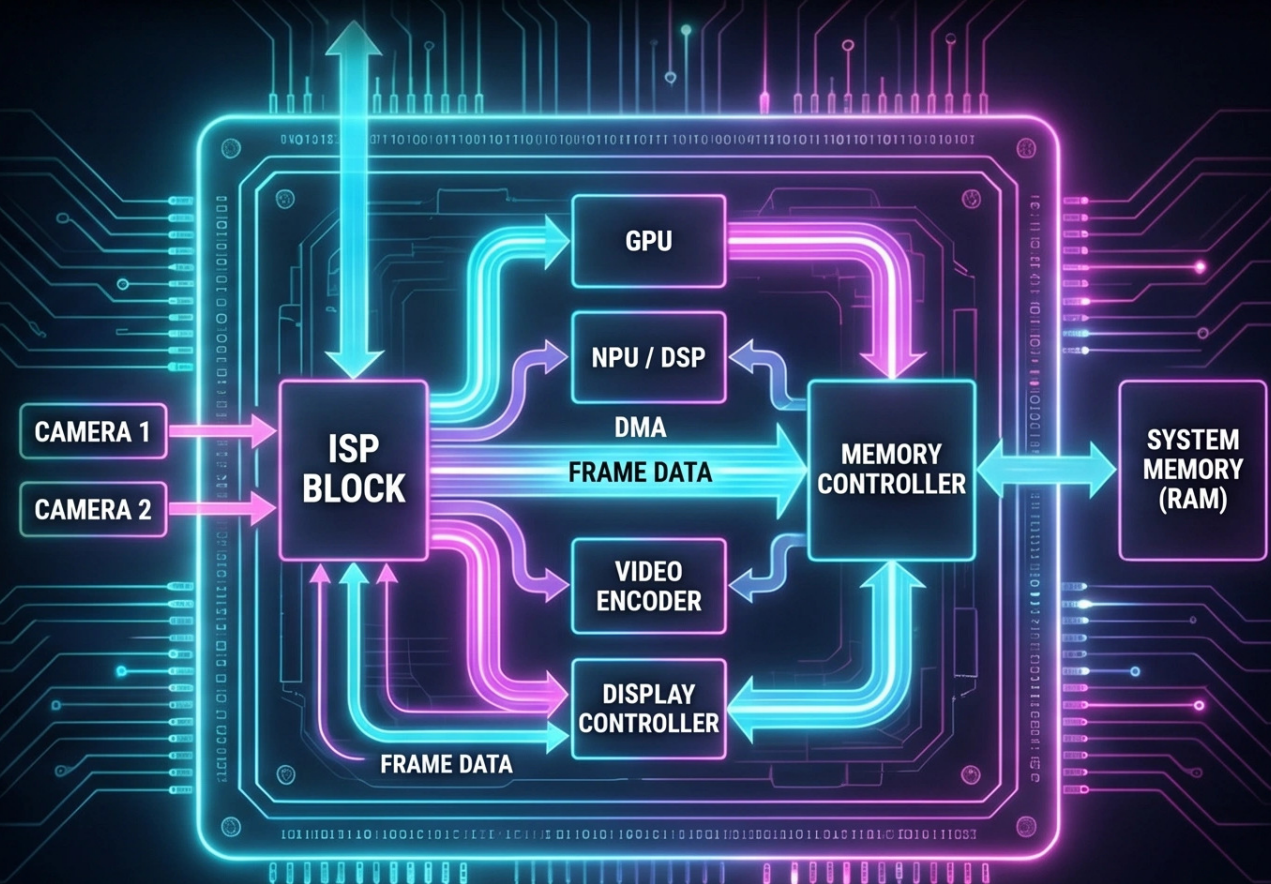


Presenter: Engineering Lecture Series

ISP TO SoC: SYSTEM INTEGRATION ARCHITECTURE

Understanding the Complete Camera Pipeline – Sensor, ISP, SoC, and Software Stack

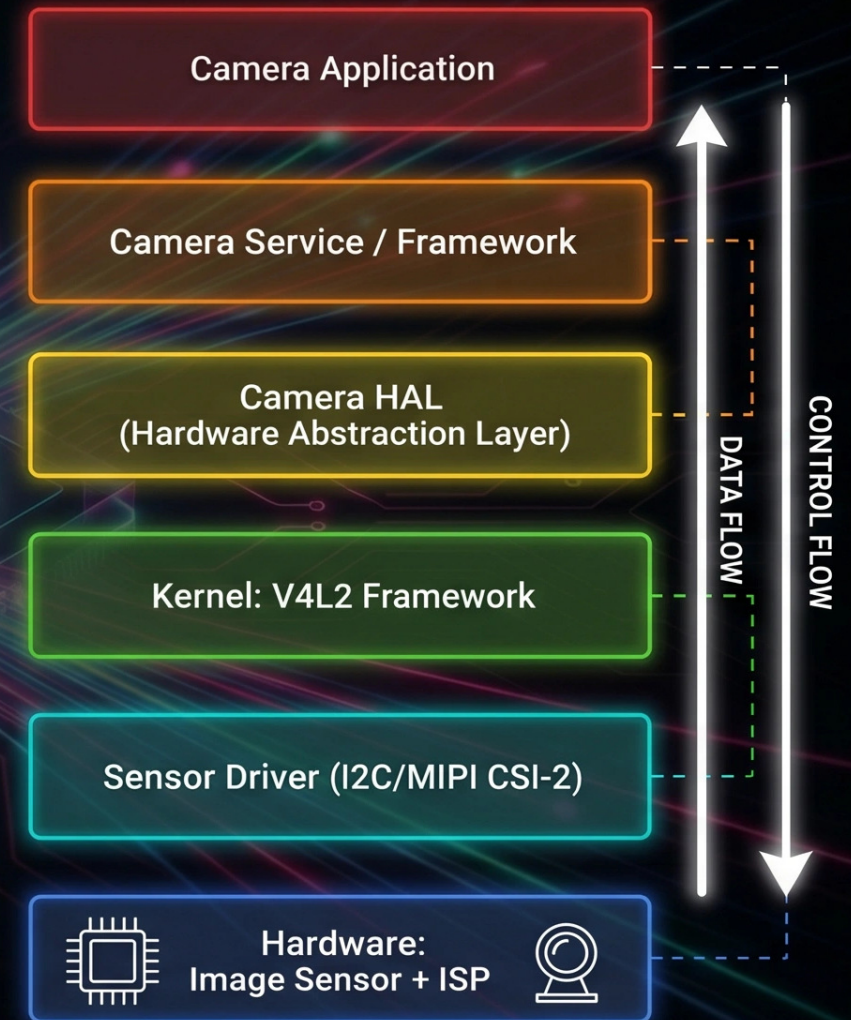
- The ISP is typically integrated into the Application Processor SoC (e.g., Snapdragon, Apple A-series).
- ISP output feeds into multiple SoC subsystems: video encoder, display controller, GPU, and NPU/DSP.
- Modern SoCs include dual ISPs for simultaneous processing of multiple cameras.
- DMA (Direct Memory Access) transfers frame data between ISP, memory, and other blocks without CPU intervention.



CAMERA SOFTWARE STACK: FROM KERNEL TO APP

KEY POINTS

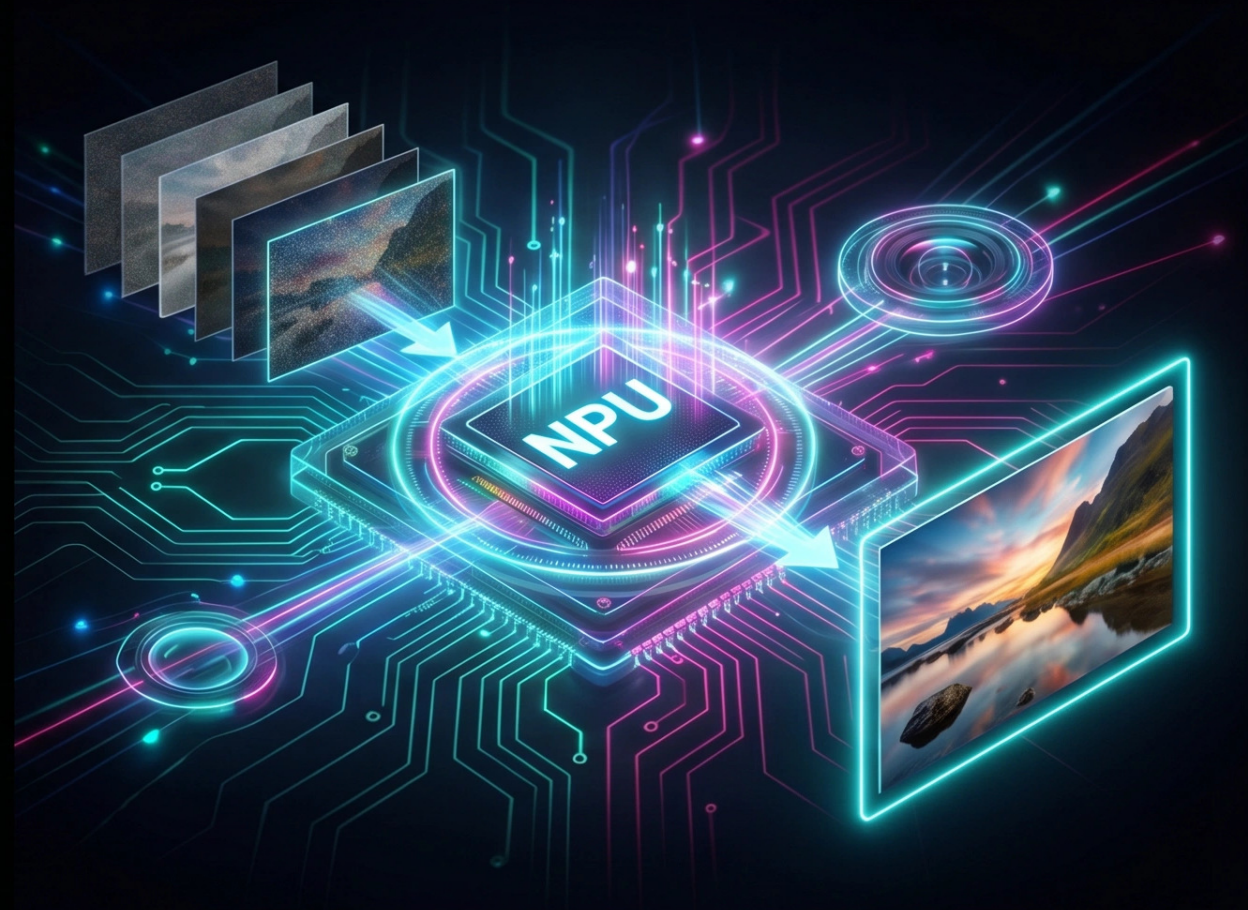
- 1. Layer 1 – Sensor Driver:** Kernel module configuring sensor via I2C and managing MIPI CSI-2.
- 2. Layer 2 – V4L2 Framework:** Linux kernel subsystem providing standardized API.
- 3. Layer 3 – Camera HAL:** Hardware Abstraction Layer translating ISP controls.
- 4. Layer 4 – Camera Service/Framework:** Manages sessions and buffers (e.g., Android Camera2).
- 5. Layer 5 – Application:** The camera app initiating capture and presenting viewfinder.



MODERN COMPUTATIONAL PHOTOGRAPHY PIPELINE

Leveraging AI, Multi-Frame Processing, and Advanced Hardware for Superior Imaging.

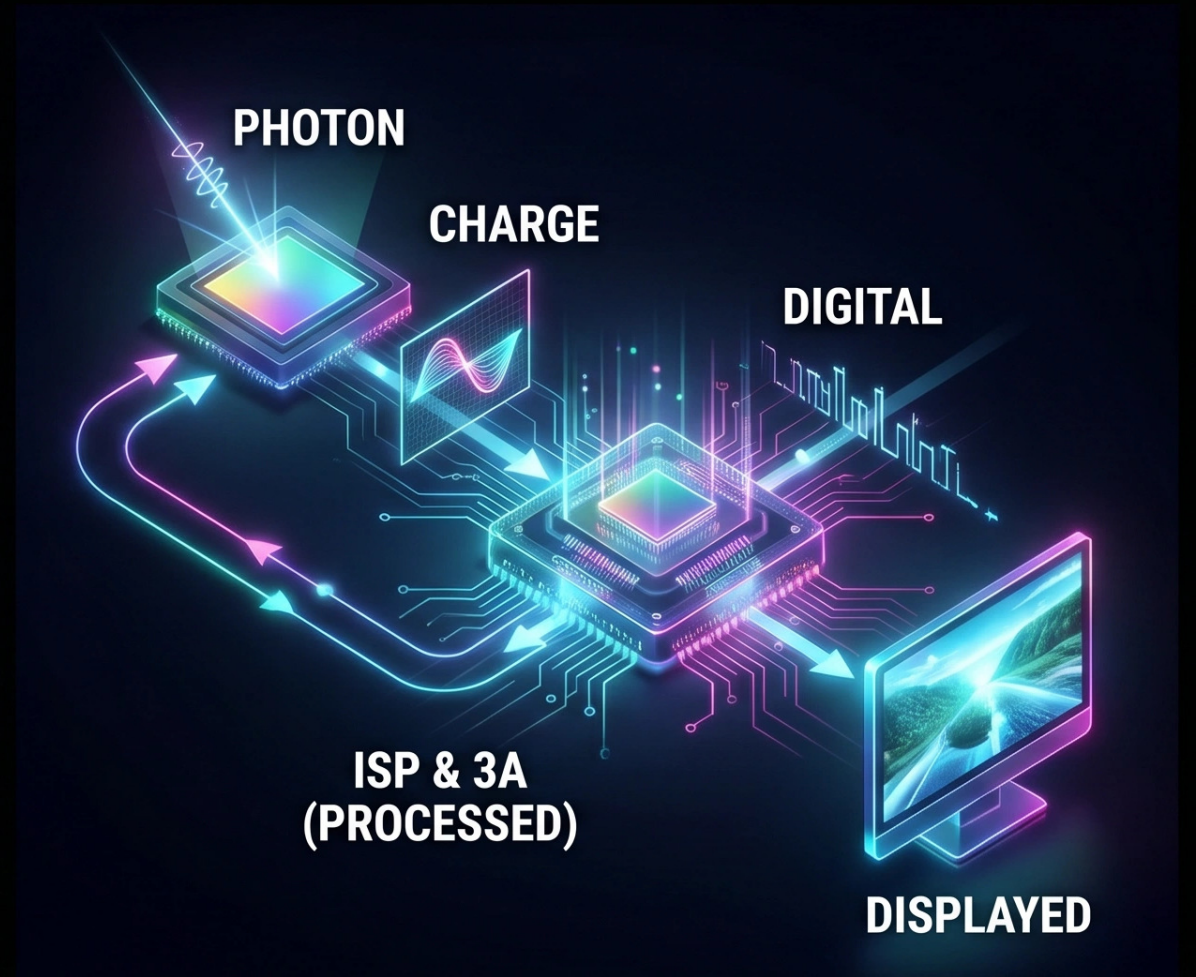
- **Beyond traditional ISP:** Modern cameras leverage GPU, DSP, and NPU for advanced processing.
- **Multi-frame HDR:** Captures multiple exposures and merges them for extended dynamic range.
- **Night Mode:** Aligns and stacks dozens of frames to reduce noise in extreme low light.
- **AI-based processing:** Neural networks for scene detection, semantic segmentation, bokeh rendering, and super-resolution.
- **Blurring Boundaries:** The boundary between ISP hardware and software-based processing continues to blur.



Presenter: Advanced Imaging Technologies Series

SUMMARY: THE COMPLETE CAMERA DATA PATH

- **Photon -> Charge -> Digital -> Processed -> Displayed:** The full journey takes under 33ms for 30fps video.
- The sensor captures raw single-channel data; the ISP reconstructs, corrects, and enhances the image.
- 3A algorithms close the loop between processed output and sensor control.
- The software stack abstracts hardware complexity, enabling application developers to focus on user experience.
- Understanding each stage is critical for camera system design, ISP tuning, and image quality optimization.



< 33ms Journey for 30fps Video